



The Mathematics Community in Myanmar, a Developing Country Plagued by Unrest

Michael Dorff and Graeme Fairweather

There is social and political unrest in many regions of the world right now that is affecting mathematicians and mathematics communities. Here we provide a glimpse of the situation in Myanmar. On February 1, 2021, the military arrested the democratically elected members of the government of Myanmar (formerly Burma) including Aung San Suu Kyi, the country’s civilian leader. You may recall that she was awarded the Nobel Peace Prize in 1991 while under house arrest for her non-violent efforts for democracy and human rights. Many citizens have taken to the streets to protest the coup, resulting in the loss of life of hundreds and the incarceration of thousands, and these numbers continue to rise daily. While many people may have heard of these devastating events in Myanmar, most people are unaware of Myanmar’s very active mathematics community. That community has been collaborating with its counterparts in Cambodia and Laos in a way that benefits all three participating nations.

In 2008, a series of annual conferences entitled “International Conference on Science and Mathematics Education in Developing Countries” was established by Dr. Chan Roath, President of the Cambodian Mathematical Society, and former Director of Cambodian Department of Scientific Research, Ministry of Education, Youth and Sports. The primary goal of these conferences is to promote mathematics and mathematics education in Southeast Asia with particular focus on Cambodia, Myanmar, and Laos. The first conference was held in Seam Reap, Cambodia, within a few miles of the World Heritage Site, Angkor Wat (think of a real-life version of where the monkeys live in the *Jungle Book*). Cambodia’s capital,

The sun setting over a Buddhist temple on Taungthaman Lake near the U Bein Bridge in Mandalay, Myanmar.

Organizers and main speakers at the Seventh International Conference in Mandalay, Myanmar in 2014. Graeme Fairweather is in the first row, fourth from the right, and his wife is behind his right shoulder.

Phnom Penh, was the venue for the next four conferences. In 2013, as military restrictions in Myanmar began to be relaxed, the conference venue was moved to Myanmar, and it was there, shortly after the sixth conference, the first of four held in Mandalay, that the Myanmar Mathematical Society (MMS) was launched. A month later, the MMS was formally admitted as a member of SEAMS, the Southeast Asian Mathematical Society. Within two months of its founding, MMS had over 900 members. The sixth conference was particularly noteworthy because it was the first international mathematics conference to be held in Myanmar. The eighth conference was planned for November 2015 in Yangon but was delayed due to the nation's general election that was scheduled for the same dates. At the conference a month after the election, spirits were running high following Aung San Suu Kyi's election victory.

Monita Chanroath, who is from Cambodia and just completed an undergraduate degree in math education, and Michael Dorff examining a soap film display after dipping a frame into a bucket of soap water.



To further promote mathematics to a broader audience in Myanmar, several significant activities took place at the Myanmar conferences. One was the introduction of three two-day preconference workshops for high school teachers, undergraduate students, and mathematics instructors/graduate students. From 2013–2016, these were held on a campus near the conference site, but in 2017, the venue was at the University of Taunggyi which is about 100 miles (a half-day bus ride) from Mandalay, the site of the conference. Taunggyi is the largest city in Shan State, southwest of Mandalay and is near picturesque Inle Lake, which the workshop leaders visited on their return journey to Mandalay. An iconic photo of a fisherman on the lake graces the cover of a recent textbook by Gilbert Strang and an intriguing account of the life of a Shan prince, a Colorado School of Mines alumnus, and his Austrian wife, Inge Sargent, is given in the book *Twilight over Burma, My Life as a Shan Princess*. A second activity was the meeting of the regional and international conference organizers with officials from the Myanmar Ministry of Education and the regional government. There is a lot of good that can come about from mathematicians reaching out to and talking with political leaders.

After five conferences in Myanmar, which were all directed by Prof. Dr. Nyunt Swe, Pro-Rector, Mandalay University of Distance Learning, the venue was switched to Laos where the conference was held in 2018 and 2019 in the capital Vien-





A mathematician from Myanmar and a mathematician from the U.S. at the Mahamuni Buddha Temple and Maha Myat Muni Pagoda in Mandalay, Myanmar.

tiane, on the banks of the Mekong River, the boundary between Laos and Thailand. The 2020 conference was intended to be held in Souphanouvong University in Luang Prabang, a World Heritage town, north of the capital, but because of the pandemic, it has been postponed. The Laos conferences have been directed by Dr. Sackmone Sirisack of the National University of Laos, Vientiane.

In all of the host cities, the conferences were well publicized, with banners across many main streets and posters on campus and surrounding areas. The local newspapers and television media were also involved in spreading the word of

mathematicians from across the world who were meeting in their country. Opening and closing ceremonies are invariably attended by local and national government officials. Participants from the international mathematics community have come from Australia, Bangladesh, Cambodia, China, France, India, Indonesia, Italy, Japan, Laos, Malaysia, Nepal, the Philippines, Singapore, South Korea, Sweden, Thailand, and Vietnam. Also, from the United States, Betsy and Joe Yanik from Emporia State University, Padhu Seshaiyer from George Mason University, Helene Tyler from Manhattan College, John Meakin from the University of Nebraska, and the authors have been involved.

Cambodia and Laos were very badly damaged during the Vietnam War. The growth of mathematics in Cambodia was further impacted by the actions of the Khmer Rouge under the Pol Pot regime during the late 1970s. However, there has been significant growth in its mathematics community since the 1st International Conference on Science and Mathematics Education in Developing Countries in 2008. Attendance at the initial conferences in Cambodia grew from 65 participants including 10 from seven countries in 2008 to 250 participants including 56 from 17 countries in 2012. All of the Myanmar conferences were well attended with around 400 participants from 20 different countries each year. In 2010, there were five mathematics PhDs in Cambodia of which only one was in academia; that number doubled over the following decade. Laos currently has six whereas Myanmar, the most populous of the three countries, has several hundred.

The current situation in Myanmar universities is rather grim. Universities re-opened in early May, but it appears that only a small percentage of faculty and an even smaller percentage of students have returned. Other faculty have been suspended indefinitely without pay although they were promised half-pay. While we are deeply concerned about the safety and security of our Myanmar colleagues, we also ask you to remember mathematicians throughout the U.S. and the world who are experiencing political and social injustice.

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